LAB EXPERIMENT RECORD

SET 23: EMPLOYEE PERFORMANCE ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	P. MOHITH KUMAR REDDY
Register Number	19254223
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Day	Thursday

Aim

To analyze employee performance using a department distribution bar chart, a performance line chart, an interactive dashboard, and a data table in R and Shiny.

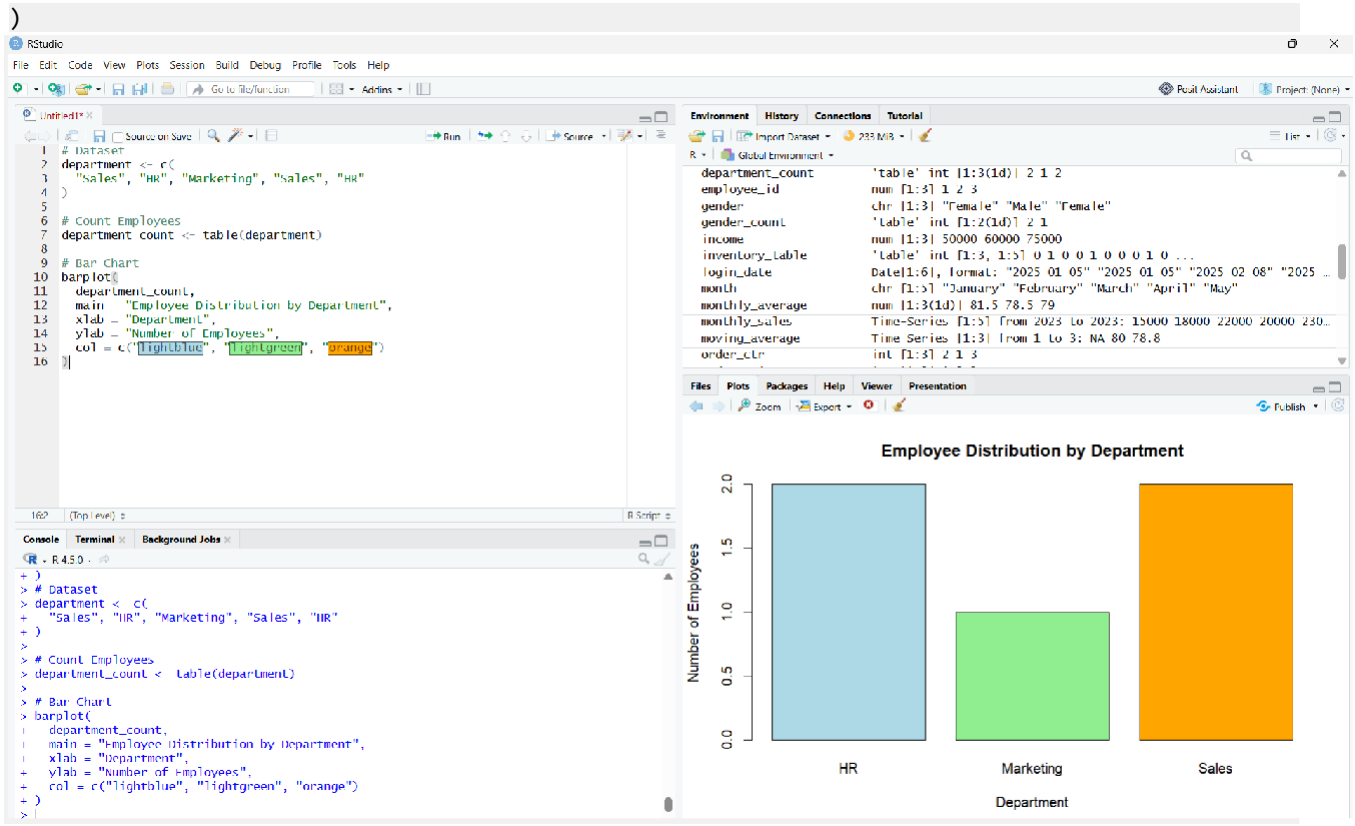
Dataset: Monthly Sales

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1: Bar Chart of Employees by Department

To visualize the distribution of employees across different departments using a bar chart.

```
# # Dataset
department <- c(
  "Sales", "HR", "Marketing", "Sales", "HR"
)
# Count Employees
department_count <- table(department)
# Bar Chart
barplot(
  department_count,
  main = "Employee Distribution by Department",
  xlab = "Department",
  ylab = "Number of Employees",
  col = c("lightblue", "lightgreen", "orange")
```



Observation: Sales and HR each have two employees, while Marketing has one employee.

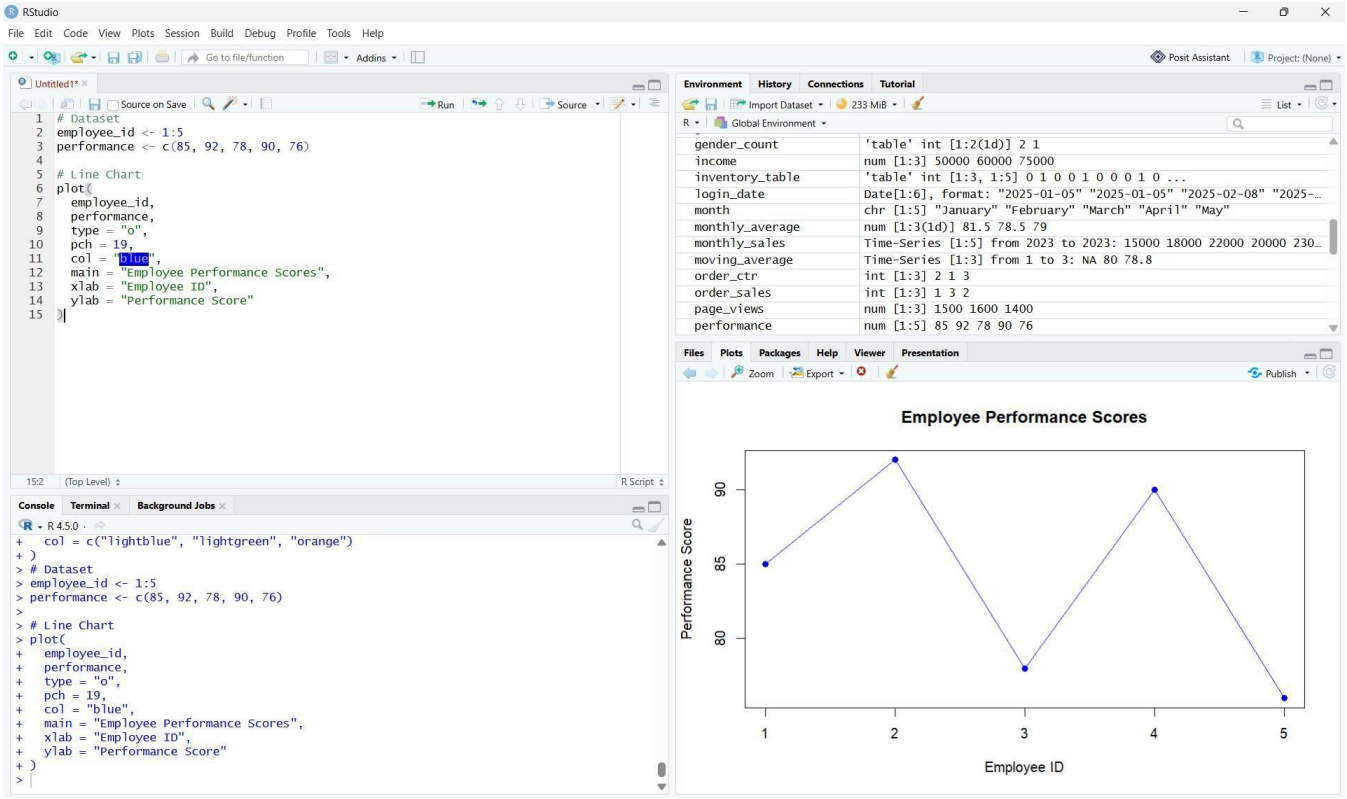
PROGRAM 2: Scatter Plot of Advertising Budget vs Monthly Sales

T o visualize employee performance scores using a line chart.

Sample Dataset

```
month <- c("January", "February", "March", "April", "May")
advertising <- c(5000, 6000, 7000, 8000, 9000)
sales <- c(15000, 18000, 22000, 20000, 23000)
# Scatter Plot
plot(
  advertising,
  sales,
  main = "Advertising Budget vs Monthly Sales",
  xlab = "Advertising Budget ($)",
  ylab = "Monthly Sales ($)",
  pch = 19,
  col = "red"
)

text(
  advertising,
  sales,
  labels = month,
  pos = 3
))
```



Observation: Employee 2 has the highest performance score of 92, while Employee 5 has the lowest score of 76.

PROGRAM 3: Interactive Monthly Sales Dashboard

To create an interactive dashboard combining a department distribution bar chart and a scatter plot of years of service versus performance scores.

```
library(shiny)


employee_id <- 1:5
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
years <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)

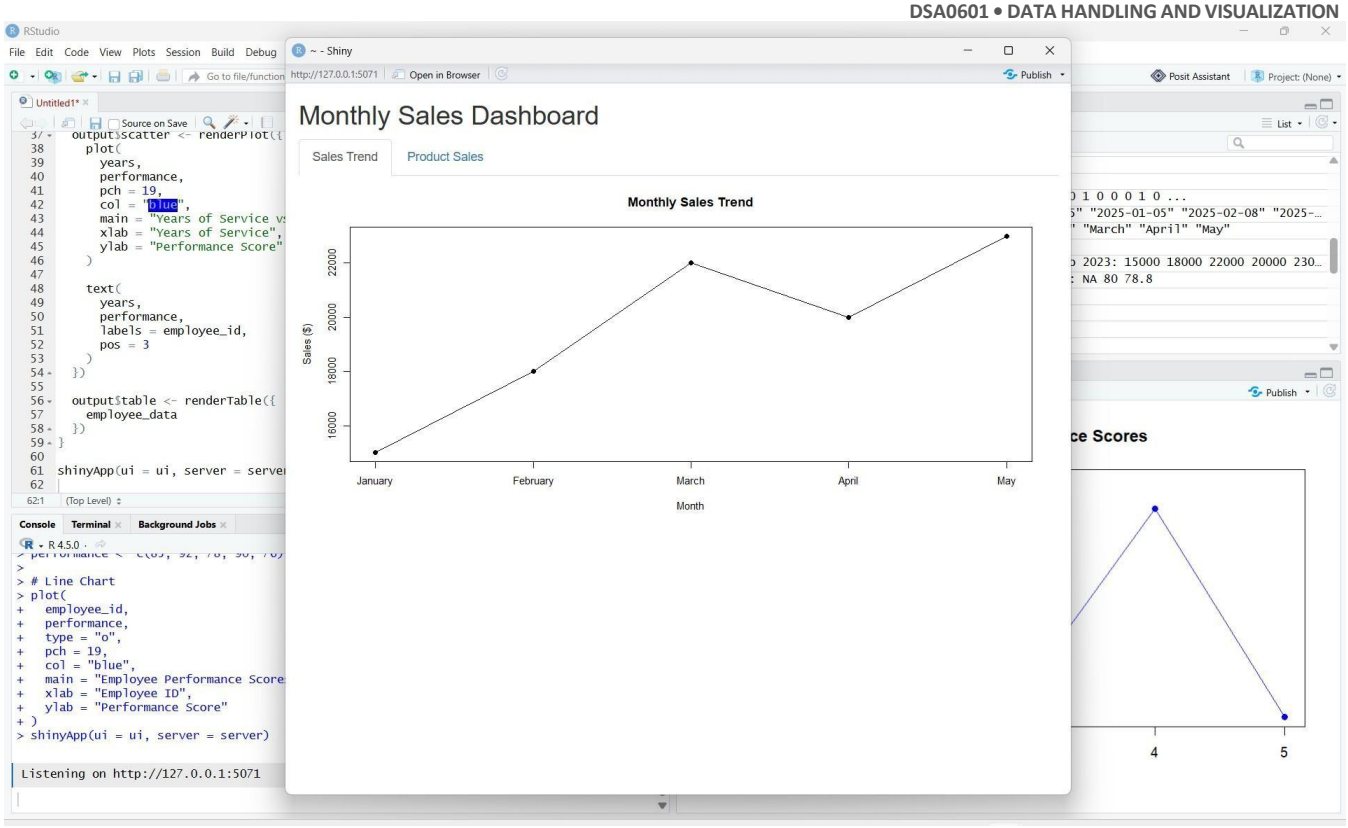

employee_data <- data.frame(
  Employee_ID = employee_id,
  Department = department,
  Years_of_Service = years,
  Performance_Score = performance
)


ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),


  tabsetPanel(
    tabPanel("Department Distribution", plotOutput("bar")),
    tabPanel("Service vs Performance", plotOutput("scatter")),
    tabPanel("Employee Data", tableOutput("table"))
  )
)


  output$table <- renderTable({
    employee_data
  })
}


shinyApp(ui = ui, server = server)shinyApp(ui = ui, server = server))
```



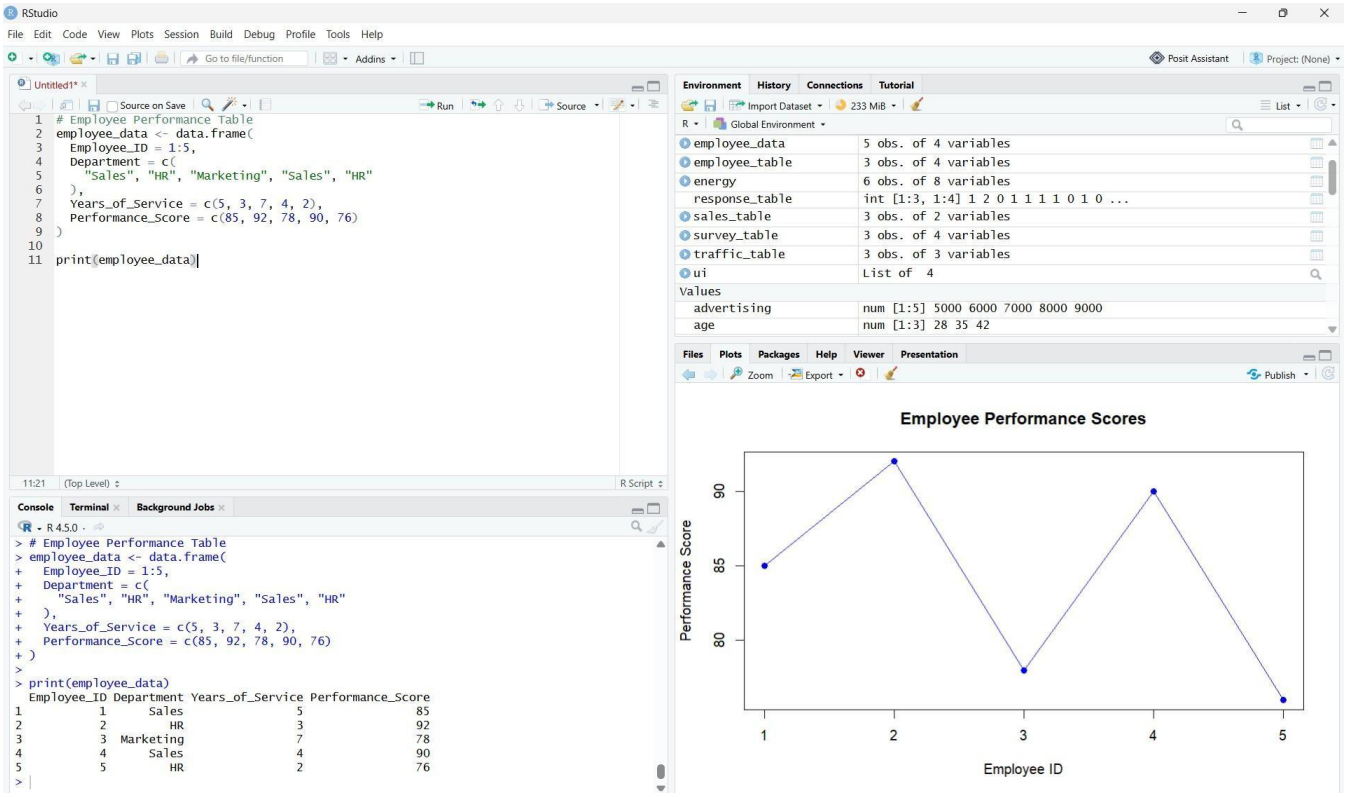
Insight: The dashboard displays department distribution and the relationship between years of service and performance scores. The employee table provides the corresponding details.

PROGRAM 4: Employee Performance Table

To display employee performance data in a structured table with appropriate column labels.

```
# # Employee Performance Table
employee_data <- data.frame(
  Employee_ID = 1:5,
  Department = c(
    "Sales", "HR", "Marketing", "Sales", "HR"
  ),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee_data)in = "Autocorrelation of Monthly Sales"
)}
```



Observation: The table displays employee IDs, departments, years of service, and performance scores.

Procedure

1. Create vectors containing employee IDs, departments, years of service, and performance scores.
2. Use table() and barplot() to visualize department distribution.

3. Use plot() to create a line chart of employee performance scores.
4. Develop a Shiny dashboard combining the department bar chart, service-versus-performance scatter plot, and employee table.
5. Create a data frame using data.frame() to display employee details.
6. Run the programs in RStudio and examine the results.

Result

Employee performance data was successfully analyzed using a bar chart, line chart, scatter plot, and table. An interactive Shiny dashboard was created to explore department distribution and employee performance.